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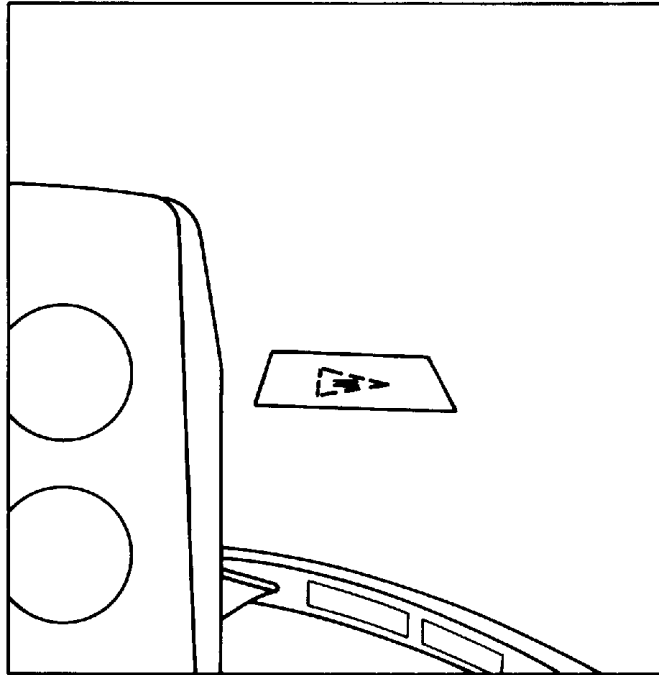


Fig. 17 VTOL - Landing Decision Point Sight Picture

5. PERFORMANCE DATA

5.1 TAKEOFF PERFORMANCE

NOTE Presented performance data apply to climbs without any significant bank angle.
Turning during climbing will reduce climb performance.

5.1.1 Maximum Takeoff Gross Mass

The maximum takeoff gross mass is determined using Fig. 9.

5.1.2 OEI Takeoff Distance (see Table 1)

Distance for standard 120 ft TDP height 250 m

5.1.3 Takeoff Flight Path Segments I and II

The OEI standard takeoff flight path (see Fig. 18) begins at the end of the “takeoff distance”, at 50 feet above the takeoff surface or higher, with positive rate of climb and V_{TOSS} , and is divided into two segments:

Segment I OEI climb through segment I has to be accomplished with V_{TOSS} and 2.5-min Power until reaching 200 ft AGL. Segment I climb performance data (Height Gain Over a Horizontal Distance of 100 ft) are presented in Fig. 5.

Segment II OEI climb through segment II has to be accomplished with V_Y and 30-min Power until reaching 1000 ft AGL. Segment II climb performance data (Height Gain Over a Horizontal Distance of 100 ft) are presented in Fig. 6.

The horizontal distance required for acceleration from V_{TOSS} to V_Y is 600 m (1980 ft) under calm wind conditions.

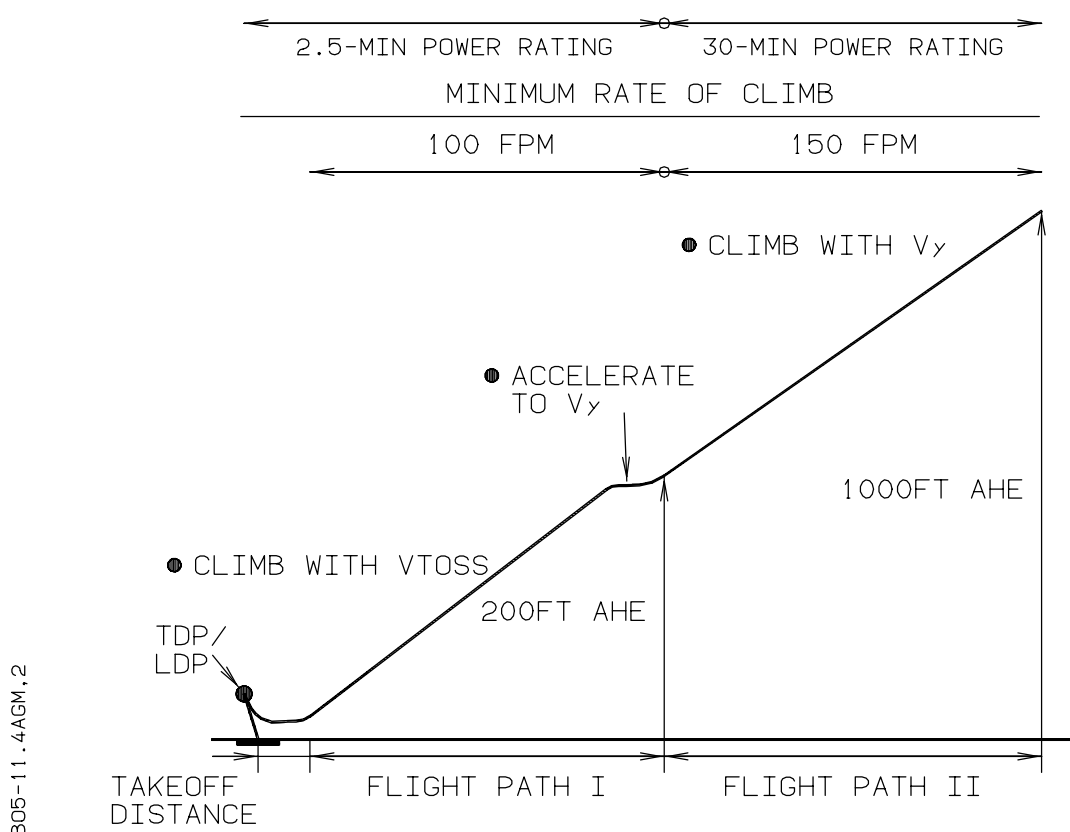


Fig. 18 VTOL - OEI Takeoff Profile

5.2 LANDING PERFORMANCE

5.2.1 OEI Landing Distance

Distance to clear a 25-ft obstacle 35 m

5.2.2 Maximum Landing Gross Mass

The maximum landing gross mass is determined using Fig. 9.

5.3 MODIFIED FLIGHT PATH TO CLEAR HIGH OBSTACLES

5.3.1 Modified Takeoff Flight Path

Some heliports may require an increase in the standard TDP height due to obstacles in the close surroundings of the site. For this purpose, the following procedure for varying the TDP standard height is permissible.

Depending on the location of the obstacle relative to the heliport, two different procedures have been established:

5.3.1.1 Distance From Heliport To Obstacle is Less Than the Takeoff Distance Required (see Fig. 19)

Establish the TDP height such that the minimum height presented in Table 1 is not lower than the obstacle height plus the minimum clearance as defined by the operational rules.

EXAMPLE: (see Fig. 19 and Table 1)

Determine: **TDP height for vertical takeoff**

Known: Obstacle height 40 ft
 Obstacle distance 200 m (in direction of departure)

Solution: 1. Add minimum clearance as defined by operational rules (e.g. 35 ft) to known obstacle height (40 ft) to obtain minimum height for takeoff (75 ft).
2. Using Table 1, select for the derived minimum height (75 ft) the corresponding TDP height = 160 ft.

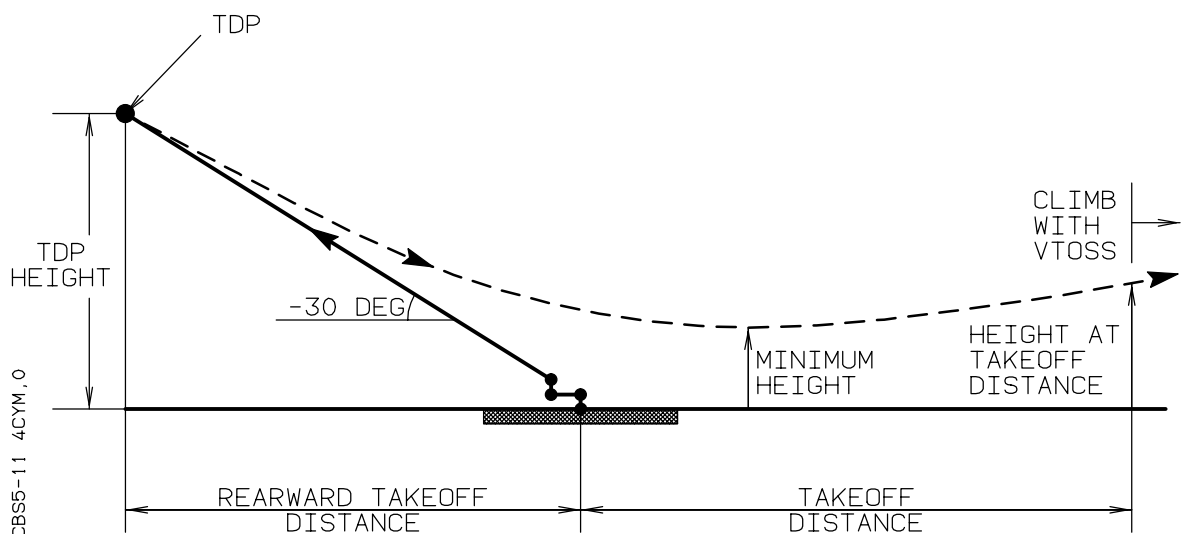


Fig. 19 VTOL - Modified Flight Path to Clear High Obstacles

TDP Height (ft)	Rearward Take-off Distance (m)	Minimum Height (ft)	Height at Takeoff Distance ¹⁾ (ft)	Maximum Take-off Distance (m)
120	76	35	50	250
140	87	55	70	240
160	98	75	90	230
180	108	95	110	220
200	119	115	130	210
¹⁾ Height at which V_{TOSS} and positive rate of climb are achieved.				

Table 1 VTOL - Distances and Heights with Variable TDP

5.3.1.2 Distance From Heliport To Obstacle is Greater Than the Takeoff Distance Required

First, from the OEI climb performance data charts (see Fig. 5 and 6), determine the climb-out height attainable over the distance from the heliport to the obstacle less the takeoff distance required. The height thus obtained plus the height for the takeoff distance shown in Table 1 is the attainable height above the takeoff surface when the helicopter reaches the obstacle. Select a TDP height such that this attainable height is not lower than the obstacle height plus the minimum clearance as defined by the operational rules. Since the takeoff distance required is reduced by an appropriate amount (see Table 1), the calculation must be repeated until an acceptable TDP height is obtained.

EXAMPLE: (see Fig. 19 and Table 1)

Determine: **TDP height for Vertical Takeoff**

Known:

Pressure Altitude	SL
OAT	15 °C
Gross Mass	2200 kg
Obstacle Height:	40 ft
Obstacle Distance:	350 m (in direction of departure)
Takeoff Distance Required	250 m (Table 1)

Solution:

- With the known MAT data (SL, 15 °C, 2200 kg), plot the climb performance using the OEI Height Gain - Segment I Chart (see Fig. 5) and calculate for a horizontal distance of 100 m (approx 11.5 ft per 100 ft, i.e. approx 38 ft).
- Calculate minimum height required at the obstacle = 82 ft:

Obstacle height	40 ft
Minimum clearance	35 ft
- Calculate minimum height required at takeoff distance (82 ft minus 38 ft, i.e. 44 ft).
- Using Table 1, round up to the next highest line (Height at Takeoff Distance - 50 ft) giving TDP height of 120 ft.

5.3.2 Modified Landing Flight Path

For varying the LDP height in order to clear obstacles when carrying out a go-around, the same procedure may be used accordingly. In this case, the speed should be increased linearly; i. e.:

200 ft/40 KIAS

150 ft/30 KIAS

100 ft/20 KIAS

NOTE Approach airspeeds can be increased by half of the wind speed.